Attention Is All You Need

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ABSTRACT

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs.

1. Introduction & Motivation

Recurrent neural networks, long short-term memory (LSTM) and gated recurrent units (GRU) have been firmly established as state of the art approaches in sequence transduction problems such as machine translation. Recurrent models factor computation along symbol positions. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths. In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on self-attention mechanisms to compute representations of its input and output without using sequence-aligned RNNs or convolution.

2. Background & Related Architectural Paradigms

The goal of reducing sequential computation forms the foundation of Extended Neural GPU, ByteNet and ConvS2S, which use convolutional networks. In these models, the number of operations required to relate signals from two arbitrary positions grows with distance: linearly for ConvS2S and logarithmically for ByteNet. In the Transformer, this is reduced to a constant number of operations O(1) via multi-head self-attention.

3. Model Architecture & Scaled Dot-Product Attention

The Transformer follows an encoder-decoder architecture using stacked self-attention and point-wise, fully connected layers. The encoder is composed of N = 6 identical layers, each with multi-head self-attention followed by a position-wise feed-forward network with residual connections and layer normalization: Output = LayerNorm(x + Sublayer(x)).
Scaled Dot-Product Attention is computed as: Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V, where queries and keys have dimension d_k = 64, and values have dimension d_v = 64. Scaling by 1 / sqrt(d_k) counteracts large magnitude dot products that push the softmax function into regions with dangerously small gradients.

3.1 Multi-Head Attention Dynamics

Instead of performing a single attention function with d_model-dimensional keys, values, and queries, we project the queries, keys, and values h = 8 times with different, learned linear projections to d_k, d_k, and d_v dimensions respectively: MultiHead(Q,K,V) = Concat(head_1, ..., head_h) * W^O. Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.

4. Experimental Results and Translation Benchmarks

On the WMT 2014 English-to-German translation task, the Transformer (Big) model achieves 28.4 BLEU, outperforming existing models and ensembles by >2.0 BLEU with an efficiency of 2.3 x 10^19 FLOPs. On WMT 2014 English-to-French, our model establishes 41.8 BLEU with training time of 3.5 days on 8 NVIDIA P100 GPUs.

5. Critical Limitations and Quadratic Scaling Constraints

Despite empirical dominance, self-attention suffers from fundamental algorithmic boundaries. First, memory and computational complexity scale quadratically O(n^2) with sequence length n. This severely bounds context windows on long texts and genomic sequences. Second, lack of recurrence requires absolute or learned positional encodings, which exhibit poor out-of-distribution length generalization on unseen token distances.

6. Open Questions & Future Research Trajectories

Research is critically needed to design sub-quadratic or linear attention mechanisms (such as state-space models and kernelized projections) and to extend attention mechanics natively to non-textual modalities without token fragmentation.